ESE 4013 Question Map

Every question from the 2020, 2021, 2022, 2024 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries.
Answers are in the ESE 4013 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 6 h for Part A and 6 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: the two numericals, overhang sizing (A12) and the sol-air wall heat gain (B4), are short and pure formula marks: learn them with the calculators even if they sit below the pass line. In Part B, B6 (transient heat flow) and B8 (degree-days) are long but come every year, so learn their diagrams and the one key equation each.

Section A: Built environment, climate, sun, shading, daylighting

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
A1 Indoor environment control and IEQ	Very likely 100%	Easy	10%
A4 Reducing energy use: new and existing buildings, lighting, building categories	Very likely 100%	Easy	18%
A11 Sun shading devices: types, functions, selection	Very likely 100%	Easy	25%
A6 Noise and acoustic comfort	Very likely 90%	Easy	30%

Block 2: very likely, medium effort · 3.8 h

Topic	Chance	Level	Covered
A7 Sun-earth relationship, solar radiation and building orientation	Very likely 100%	Medium	42%
A10 Wind flow patterns and natural ventilation around buildings	Very likely 100%	Medium	48%
A13 Daylighting systems: passive, active (sun tracker), top and side lighting	Very likely 90%	Medium	60%
A8 Ambient temperature and sky temperature	Very likely 90%	Medium	64%
A14 Daylighting design principles and building orientation/form	Very likely 80%	Medium	71%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
A12 Shadow angles (HSA, VSA) and overhang sizing numerical	Very likely 90%	Hard	76%

✔ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 6.4 h)

Block 4: likely · 0.8 h

Topic	Chance	Level	Covered
A2 Indoor air contamination: sources and control	Likely 70%	Easy	82%
A3 Energy use in buildings: envelope, use pattern, performance factors	Likely 60%	Easy	89%

Only if time is left · 1.5 h

Topic	Chance	Level	Covered
A5 Thermal and visual comfort in building design	Possible 50%	Medium	96%
A9 Urban heat island and urban wind island effects	Possible 40%	Medium	100%

Section B: Thermal performance, loads, annual energy, NZEB

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
B11 Net-zero energy and green buildings, LEED	Very likely 100%	Easy	14%
B3 Smart windows (thermochromic, photochromic, electrochromic, LC)	Very likely 100%	Easy	19%
B10 Zoning, building shape and one-zone heat balance	Very likely 90%	Easy	22%

B5 Internal heat gains: people, lights, equipment; sensible vs latent	Very likely 80%	Easy	31%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
B7 Infiltration, ventilation and the stack effect	Very likely 100%	Medium	42%
B1 Heat transfer through windows: U-factor, double pane, low-e, glazing selection	Very likely 100%	Medium	52%
B12 Evaporative cooling and solar-assisted desiccant dehumidification	Very likely 90%	Medium	60%

Block 3: very likely numericals · 2.0 h

Topic	Chance	Level	Covered
B8 Annual energy: balance point, degree-day and bin methods	Very likely 100%	Hard	70%
B6 Transient heat flow: diurnal wall response, CTF, finite difference	Very likely 100%	Hard	80%

✔ Pass line: 80% of Section B marks covered, enough to choose 3 of the 4 questions (about 5.9 h)

Block 4: likely · 1.8 h

Topic	Chance	Level	Covered
B2 Solar heat gain coefficient (SHGC)	Likely 70%	Medium	83%
B4 Sol-air temperature, wall colour, and the sun-lit wall numerical	Likely 70%	Hard	93%

Only if time is left · 1.2 h

Topic	Chance	Level	Covered
B13 Energy audit and building energy management economics	Possible 50%	Easy	96%
B9 Below-grade heat loss: basement walls, floors, slab on grade	Possible 50%	Medium	100%

Pass plan

2021, 2022 and 2024 papers: 210 marks, 4 questions of 35 per section, answer 3 per section. 2020 was a short 120-mark paper (3 questions of 30 per section, answer 2). Section A is Kazi Sir / Hasan Ali Sir's part (introduction, climate, sun, wind, shading, daylighting). Section B is Nafisa Lubaba Ma'am's part (windows, sol-air, internal gains, transient, infiltration, annual energy, NZEB). "Must study" topics came up in 3 or 4 of the 4 papers.

SECTION A: BUILT ENVIRONMENT, CLIMATE, SUN, SHADING, DAYLIGHTING

13 topics

cover 96% of all Section A marks asked

- IEQ, indoor air contamination, energy use factors
- Sun-earth relation, sky temperature, wind patterns
- Shading devices + HSA/VSA overhang numerical
- Daylighting systems and design principles

SECTION B: THERMAL PERFORMANCE, LOADS, ANNUAL ENERGY, NZEB

10 topics

cover 89% of all Section B marks asked

- Glazing, low-e, SHGC, smart windows
- Sol-air temperature + wall numerical, internal gains
- Transient heat flow (CTF, finite difference), infiltration and stack effect
- Degree-day/bin method, NZEB, desiccant cooling

- **Never skip the two numericals.** The sol-air wall heat gain (2021 5d, 13 marks) and the overhang sizing at Fulbarigate (2024 3c) are easy marks once you know two formulas each.
- **Section A is mostly definitions plus sketches.** IEQ, sun-earth relation, shading devices, daylighting systems and wind patterns repeat every year with slightly different wording.
- **Section B repeats the same eight ideas:** glazing and low-e, SHGC, smart windows, sol-air, internal gains, transient/CTF/finite difference, infiltration/stack effect, degree-day and bin, NZEB and desiccant cooling.
- **"Good to have"** topics are the backup for your third question in a section.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: BUILT ENVIRONMENT, CLIMATE, SUN, SHADING, DAYLIGHTING: MARKS ASK

Topic	2024	2022	2021	2020	Years	Marks	
A1 Indoor environment control and IEQ	11	8	11	20	4/4	50	Must study
A4 Reducing energy use: new and existing buildings, lighting, building categories	8	17	7	8	4/4	40	Must study
A11 Sun shading devices: types, functions, selection	12	7	12	5	4/4	36	Must study
A6 Noise and acoustic comfort	8	8	8		3/4	24	Must study
A7 Sun-earth relationship, solar radiation and building orientation	10	10	19	22	4/4	61	Must study
A10 Wind flow patterns and natural ventilation around buildings	7	7	10	6	4/4	30	Must study
A13 Daylighting systems: passive, active (sun tracker), top and side lighting	25	10	23		3/4	58	Must study
A8 Ambient temperature and sky temperature	5	8	6		3/4	19	Must study
A14 Daylighting design principles and building orientation/form	10	8		16	3/4	34	Must study
A12 Shadow angles (HSA, VSA) and overhang sizing numerical	8	20			2/4	28	Must study
A2 Indoor air contamination: sources and control	8		12	10	3/4	30	Must study
A3 Energy use in buildings: envelope, use pattern, performance factors	8		24		2/4	32	Must study
A5 Thermal and visual comfort in building design		29	8		2/4	37	Must study
A9 Urban heat island and urban wind island effects	20				1/4	20	Good to have

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: THERMAL PERFORMANCE, LOADS, ANNUAL ENERGY, NZEB: MARKS ASKED

Topic	2024	2022	2021	2020	Years	Marks	
B11 Net-zero energy and green buildings, LEED	7	19	26	20	4/4	72	Must study
B3 Smart windows (thermochromic, photochromic, electrochromic, LC)	7		9	8	3/4	24	Must study
B10 Zoning, building shape and one-zone heat balance	7	8	5		3/4	20	Good to have
B5 Internal heat gains: people, lights, equipment; sensible vs latent	17	16		11	3/4	44	Must study
B7 Infiltration, ventilation and the stack effect	9	27	8	12	4/4	56	Must study
B1 Heat transfer through windows: U-factor, double pane, low-e, glazing selection	17	14	18	5	4/4	54	Must study
B12 Evaporative cooling and solar-assisted desiccant dehumidification	22	8	9		3/4	39	Must study
B8 Annual energy: balance point, degree-day and bin methods	14	13	16	10	4/4	53	Must study
B6 Transient heat flow: diurnal wall response, CTF, finite difference	18	17	8	7	4/4	50	Must study
B2 Solar heat gain coefficient (SHGC)	11	9			2/4	20	Must study
B4 Sol-air temperature, wall colour, and the sun-lit wall numerical	11		20	17	3/4	48	Must study
B13 Energy audit and building energy management economics		8	11		2/4	19	Good to have
B9 Below-grade heat loss: basement walls, floors, slab on grade		9	10		2/4	19	Good to have

Section A: Built environment, climate, sun, shading, daylighting

questions 1-4 (1-3 in 2020)

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A1 Indoor environment control and IEQ

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 50 marks in total

STUDY THESE

- IEQ = IAQ + thermal + visual (illumination) + acoustic: the four aspects
- Internal and external factors on which building energy use depends
- Adverse effects of poor IEQ (sick building symptoms) and how ventilation / good housekeeping help

> 2024 Q1(b) 11 marks
>
> What is meant by IEQ? What are the adverse effects of poor IEQ? How good housekeeping can provide good IEQ?

> 2022 Q1(a) 8 marks
>
> What is meant by 'Indoor Environment control'? To control indoor environment, provide some technical suggestions with examples.

> 2021 Q1(c) 11 marks
>
> What is indoor environment control (IEQ) of a building? What are the internal and external factors upon which energy consumption of a building depend? What are the adverse effects of IEQ?

> 2020 Q1(a) 10 marks
>
> What is meant by Indoor Environment Control of a building? What are the internal and external factors upon which energy use of a building depend?

> 2020 Q1(b) 10 marks
>
> What are the four major aspects of indoor environment quality? How can ventilation contribute to maintain good indoor air quality?

Full answer: ESE 4013 Study Pack, topic A1.

A4 Reducing energy use: new and existing buildings, lighting, building categories

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 40 marks in total

STUDY THESE

- Seven approaches: reduce heating, cooling, ventilation, lighting, hot water, equipment energy + good housekeeping
- Key design factors for a new building (orientation, envelope, glazing, shading, daylight, natural ventilation, efficient systems, renewables)
- Building categories by occupancy (ICC groups A, B, E, F, H, I, M, R, S, U) with energy-saving options for each

> 2024 Q1(d) 8 marks
>
> When designing a new building, what key factors should be considered to minimize dependence on artificial energy consumption?

> 2022 Q1(d) 10 marks
>
> Based on indoor activities, categories the building and mention the energy saving options for each category.

> 2022 Q2(a) 7 marks
>
> For an existing building, discuss the technical issues to reduce energy consumption. Explain the strategies to ensure good acoustic environment in building.

> 2021 Q1(d) 7 marks
>
> When developing a new building, what are the factors should be considered to reduce artificial energy consumption?

> 2020 Q2(c) 8 marks
>
> How can energy use for lighting be reduced?

Full answer: ESE 4013 Study Pack, topic A4.

A11 Sun shading devices: types, functions, selection

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 36 marks in total

STUDY THESE

- Internal, external, interpane; horizontal (overhang), vertical fins, egg-crate, vegetation
- Selection by orientation: south/equator-facing horizontal, east/west vertical or movable, north little
- Functions, advantages and disadvantages; design factors (type, dimension, category: fixed/movable)

> 2024 Q3(b) 12 marks
>
> Select appropriate shading devices for different sides of a building and mention the reasons behind their selection.

> 2022 Q3(d) 7 marks
>
> Discuss about the factors that should be considered to design and select a sun-shading device.

> 2021 Q4(c) 12 marks
>
> What are the advantages and disadvantages of sun shading devices? Explain the horizontal and vertical shadow angle to determine the dimension of shading device.

> 2020 Q3(b) 5 marks
>
> What are the functions of sun shading devices? Describe how to select a sun shading device for a building.

Full answer: ESE 4013 Study Pack, topic A11.

A6 Noise and acoustic comfort

Must study · Very likely · 90% · Easy

Asked in 3 of 4 papers · 24 marks in total

STUDY THESE

- Noise definition; acoustic comfort / acoustic satisfaction
- Two tools: sound insulation (envelope, partitions) and sound absorption (room acoustics)
- Design suggestions and absorptive materials; parameters (noise level, reverberation time, STC/NRC)

> 2024 Q1(c) 8 marks
>
> What is noise? Briefly describe various methods of maintaining acoustic comfort in built environment.

2022 Q2(b) 8 marks

What is meant by acoustic comfort? Discuss about different building materials that can be used to maintain good acoustic comfort.

2021 Q2(c) 8 marks

What is meant by acoustic comfort? Explain the parameters that should be maintained for good acoustic condition in building.

Full answer: ESE 4013 Study Pack, topic A6.

A7 Sun-earth relationship, solar radiation and building orientation

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 61 marks in total

STUDY THESE

- Rotation (24 h, day/night) and revolution (365¼ days, seasons) with 23.5° tilt; solstice and equinox dates
- Weather vs climate; G = B + D + R at ground level and the factors that change it
- Orientation of a building and its surfaces: long axis east-west, south face shaded, tilt angle

2024 Q2(a) 10 marks

Write short notes on (i) Solstice, and (ii) Equinox. Describe briefly about various factors that influence the variation of radiation at ground level.

2022 Q3(a) 10 marks

On the basis of energy consumption view point, signify the sun-earth relationship for a building.

2021 Q3(a) 8 marks

Distinguish between weather and climate. Explain the effects of building orientation on the solar heat gain of a building.

2021 Q3(b) 11 marks

What does mean by sun-earth relationship? With the help of necessary sketch, explain the significance of earth relation and revolution on building heat gain.

2020 Q2(a) 10 marks

> What are meant by earth rotation and earth revolution? Explain with necessary figure.

> 2020 Q2(b) 12 marks
>
> How is built environmental influenced by the orientation of a building? Explain.

Full answer: ESE 4013 Study Pack, topic A7.

A10 Wind flow patterns and natural ventilation around buildings

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 30 marks in total

STUDY THESE

- Why control wind; windward/leeward, wind shadow (3-4 times building size)
- Five patterns: gap in windbreak, bar effect 1.4×, Venturi 1.3-1.6×, gap effect 1.2-1.5×, corner effect 1.2-2.2×, wake 1.4-2.2×
- Design modifications for natural ventilation: orientation, inlet/outlet, cross ventilation, stack, wing walls

> 2024 Q3(d) 7 marks
>
> Mention some design modifications that can influence the natural ventilation of a building.

> 2022 Q4(d) 7 marks
>
> How to increase natural wind flow around building? Discuss with schematically.

> 2021 Q3(d) 10 marks
>
> Draw five wind flow patterns around a building which will increase the natural wind flow and mention the amount of increment.

> 2020 Q3(a) 6 marks
>
> Why is control of wind motion necessary for a building? Draw four wind flow patterns around a building which can help to increase air flow.

Full answer: ESE 4013 Study Pack, topic A10.

A13 Daylighting systems: passive, active (sun tracker), top and side lighting

Must study · Very likely · 90% · Medium

Asked in 3 of 4 papers · 58 marks in total

STUDY THESE

- Daylighting definition; passive vs active; sun tracker working principle
- Top lighting (skylight, monitor, saw-tooth, clerestory, tubular) vs side lighting; light shelf / daylight redirection
- Benefits (15-40 % savings) and drawbacks; glare: causes and effects

> 2024 Q4(a) 10 marks
>
> Between active and passive day lighting systems, which one is preferable and why? Write down working principle of sun-tracker system with schematic.

> 2024 Q4(b) 9 marks
>
> Among different passive day lighting system, which one is more suitable for building in Bangladesh? Explain with diagram and examples.

> 2024 Q4(c) 6 marks
>
> What is glare? Explain its causes and effects on visual comfort.

> 2022 Q4(b) 10 marks
>
> With the help of neat sketch, describe the active daylighting system.

> 2021 Q4(a) 8 marks
>
> What are the different options of natural lighting in building? Among top and side daylighting, which one is preferable? Explain.

> 2021 Q4(b) 6 marks
>
> What is daylighting? What are the benefits and drawbacks of daylighting?

> 2021 Q4(d) 9 marks
>
> Write short notes on: (i) Saw tooth lighting, (ii) Tubular daylighting, (iii) Daylight reduction [redirection].

Full answer: ESE 4013 Study Pack, topic A13.

A8 Ambient temperature and sky temperature

Must study · Very likely · 90% · Medium

Asked in 3 of 4 papers · 19 marks in total

STUDY THESE

- Ambient temperature = air temperature in a shaded, ventilated enclosure

- Sky temperature: $T_{sky} = T_a - 6$ or Swinbank $T_{sky} = 0.0552\ T_a^{1.5}$; full clear/cloudy model
- Night-time radiative heat loss to the cold sky (roofs, radiative cooling, condensation)

> 2024 Q2(d) 5 marks
>
> Define ambient temperature and sky temperature. Explain how the difference between these two affects heat loss from building surfaces during nighttime.

> 2022 Q3(c) 8 marks
>
> How to calculate sky temperature? How sky temperature influences the analysis of solar energy utilization in building?

> 2021 Q3(c) 6 marks
>
> What is the relationship between ambient temperature and sky temperature? How sky temperature can be calculated?

Full answer: ESE 4013 Study Pack, topic A8.

A14 Daylighting design principles and building orientation/form

Must study · Very likely · 80% · Medium

Asked in 3 of 4 papers · 34 marks in total

STUDY THESE

- Six principles: orientation, form, glazing ratio and specification, window height and location, overhead daylighting, redirection
- Glazing ratio 25-50 %; roof lights max 12 % of floor area; higher window head = deeper daylight
- Best orientation and form for Bangladesh: long axis east-west, main glazing north/south, thin plan/courtyard

> 2024 Q4(d) 10 marks
>
> What should be the orientations and forms of the building according to the location of Bangladesh? Describe with schematic diagram.

> 2022 Q4(c) 8 marks
>
> What factors should be considered while designing a natural daylighting system? Explain with proper example.

> 2020 Q3(c) 16 marks

> Classify the daylighting system. In contrast of daylighting, write short note on (i) Glazing specification, (ii) Window height and location, (iii) Tubular daylight devices.

Full answer: ESE 4013 Study Pack, topic A14.

A12 Shadow angles (HSA, VSA) and overhang sizing numerical

Must study · Very likely · 90% · Hard

Asked in 2 of 4 papers · 28 marks in total

STUDY THESE

- HSA = wall azimuth − solar azimuth (for vertical fins); tan VSA = tan(altitude)/cos(HSA) (for overhangs)
- Overhang depth = window height / tan VSA; side extension from HSA
- Five design steps: cut-off date, start/end time, sun position from sun-path, HSA/VSA, depth and width

> 2024 Q3(c) 8 marks
>
> Suppose a horizontal overhang is located at the south window of a building. The building is located at Fulbarigate facing due south. The height of the window from the sill is 0.9 m and the window is 0.5 m wide. Calculate the depth and width of the overhang considering the critical month and time.

> 2022 Q3(b) 10 marks
>
> What are meant by horizontal shadow angle and vertical shadow angle? How to calculate them and mention the significance of them.

> 2022 Q4(a) 10 marks
>
> How to identify the size of shading devices? Discuss with the help of sketchmatic diagram.

> 2021 Q4(c) 0 marks
>
> (second half, marks counted in A11) Explain the horizontal and vertical shadow angle to determine the dimension of shading device.

Full answer: ESE 4013 Study Pack, topic A12.

A2 Indoor air contamination: sources and control

Must study · Likely · 70% · Easy

Asked in 3 of 4 papers · 30 marks in total

STUDY THESE

- Five source groups: outside, equipment (HVAC + non-HVAC), human activities, building components/furnishings, other
- How the HVAC system itself contaminates the building and how to prevent it
- Personal responsibility: smoking, cooking, cleaning, maintenance habits

> 2024 Q3(a) 8 marks
>
> What are the ways by which built environment can be contaminated by HVAC system? How to prevent that?

> 2021 Q2(b) 12 marks
>
> What are the major sources of indoor air contaminations? Describe the personal responsibility to reduce the indoor air contaminations.

> 2020 Q1(c) 10 marks
>
> What are the main sources of indoor air contamination?

Full answer: ESE 4013 Study Pack, topic A2.

A3 Energy use in buildings: envelope, use pattern, performance factors

Must study · Likely · 60% · Easy

Asked in 2 of 4 papers · 32 marks in total

STUDY THESE

- Building envelope (5 elements) vs building fabric; why building energy is a concern (40-50 % of energy, ~19-39 % of CO2)
- Energy use by life stage: embodied, operational, maintenance, demolition; end-use split (HVAC 35 %, lighting 11 %, appliances 18 %)
- Six factors of energy use (climate, envelope, systems, O&M, IEQ, occupants) + social factor; energy flow (gross need, gains, delivered, losses)

> 2024 Q1(a) 8 marks
>
> Define building envelop and building fabric? Why building energy is a concern of study in current times?

> 2021 Q1(a) 8 marks
>
> How should building energy performance contribute to reduce the energy consumption?

> 2021 Q1(b) 9 marks
>
> Explain the energy use in building. What are the activities influence the energy use in buildings?

> 2021 Q2(d) 7 marks
>
> Explain the energy used pattern in a building.

Full answer: ESE 4013 Study Pack, topic A3.

A5 Thermal and visual comfort in building design

Must study · Possible · 50% · Medium

Asked in 2 of 4 papers · 37 marks in total

STUDY THESE

- Thermal comfort (ASHRAE definition), six factors: 4 environmental + 2 personal; met and clo
- Visual comfort definition and lighting design considerations (task lighting, local control, no over-illumination)
- Design of walls and roofs for hot-humid, hot-dry, cold and composite climates; effect of wall/roof colour and material on visual comfort

> 2022 Q1(b) 8 marks
>
> What is meant by thermal comfort? Explain the impact of thermal control on building design.

> 2022 Q1(c) 9 marks
>
> Discuss the aspects of buildings design on the followings: (i) thermal comfort, (ii) visual comfort, (iii) acoustic comfort.

> 2022 Q2(d) 12 marks
>
> Discuss the design considerations of a building wall and roof for different climate conditions. Also discuss the effect of wall and roof materials on visual comfort.

> 2021 Q2(a) 8 marks
>
> What is visual comfort? Explain the design consideration for visual comfort of a building.

Full answer: ESE 4013 Study Pack, topic A5.

A9 Urban heat island and urban wind island effects

Good to have · Possible · 40% · Medium

Asked in 1 of 4 papers · 20 marks in total

STUDY THESE

- Urban heat island: causes chain (anthropogenic heat, less green, dark surfaces, canyons) with the slide diagram
- Urban wind island: city vs rural boundary layer; street canyon effect
- How urban geometry changes wind speed, turbulence and direction

> 2024 Q2(b) 10 marks
>
> Define urban heat island effect. With necessary diagram, briefly describe how urban heat island effect occurs.

> 2024 Q2(c) 10 marks
>
> Wind is urban areas behaves differently compared to rural areas. Discuss the urban wind island effect and evaluate how urban geometry alters wind speed, turbulence, and flow direction.

Full answer: ESE 4013 Study Pack, topic A9.

Section B: Thermal performance, loads, annual energy, NZEB

questions 5-8 (4-6 in 2020)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B11 Net-zero energy and green buildings, LEED

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 72 marks in total

STUDY THESE

- US DOE ZEB vs EU nearly-ZEB; net-zero site / source / cost / emission
- Why NZEB; design principles: envelope, efficiency, renewables (+ commissioning)

- Green building features, LEED functions and how to get certified; challenges of low-energy technologies

> 2024 Q8(b) 7 marks
>
> List and describe key sustainability features of green buildings. Describe HVAC optimization strategies for net-zero buildings.

> 2022 Q8(a) 12 marks
>
> Why net-zero energy building important? What energy efficiency measures should be taken in the building? How to design a green building?

> 2022 Q8(c) 7 marks
>
> What are the functions of LEED? How to achieve a LEED certificate? Discuss.

> 2021 Q8(a) 9 marks
>
> What is meant by nearly zero-energy building? Distinguish between net-zero site energy and net-zero source energy building.

> 2021 Q8(b) 9 marks
>
> Discuss the principles for designing a net-zero energy building.

> 2021 Q8(c) 8 marks
>
> Discuss about the challenges of technologies for low energy building.

> 2020 Q6(b) 10 marks
>
> What is the basic difference between USA and European Union on the definition of zero-energy building? Why we need net-zero energy building?

> 2020 Q6(c) 10 marks
>
> What are the principles of net-zero energy building design? Explain.

Full answer: ESE 4013 Study Pack, topic B11.

B3 Smart windows (thermochromic, photochromic, electrochromic, LC)

Must study · Very likely · 100% · Easy

Asked in 3 of 4 papers · 24 marks in total

STUDY THESE

- Smart / dynamic glazing changes optical properties with temperature, light or voltage
- Four types and how each works

- Photochromic vs thermochromic for Bangladesh (hot-humid: thermochromic)

> 2024 Q5(d) 7 marks
>
> Between photochromic and thermochromic window which one is more suitable for Bangladesh and why?

> 2022 Q5(c) 0 marks
>
> (second half, marks counted in B1) Write short note on "smart window".

> 2021 Q6(c) 9 marks
>
> What is smart window? Describe the various types of smart windows.

> 2020 Q4(d) 8 marks
>
> Write a short note on "smart window".

Full answer: ESE 4013 Study Pack, topic B3.

B10 Zoning, building shape and one-zone heat balance

Good to have · Very likely · 90% · Easy

Asked in 3 of 4 papers · 20 marks in total

STUDY THESE

- Zone = space with its own thermostat; why zoning (different loads, orientation, schedules, temperatures)
- $K_{tot} = \Sigma UA + \rho c_p \dot{V}$; heat balance; unconditioned attached spaces
- Compact square vs long narrow plan: surface-to-volume ratio

> 2024 Q6(c) 7 marks
>
> A new residential building is being designed with two floor-plan options of the same total floor area: a compact square layout and a long narrow layout. Which option will have higher heat loss, and for what reason? Propose modifications to reduce this heat loss.

> 2022 Q6(d) 8 marks
>
> What does mean by zoning of building? Why it is required? Explain with example.

> 2021 Q5(c) 5 marks
>
> Why zoning of building is required?

Full answer: ESE 4013 Study Pack, topic B10.

B5 Internal heat gains: people, lights, equipment; sensible vs latent

Must study · Very likely · 80% · Easy

Asked in 3 of 4 papers · 44 marks in total

STUDY THESE

- Components: occupants (sensible + latent), lighting, equipment/motors, (plus infiltration latent)
- Lighting: $Q = P_L F_{UL} F_{SA}$ or LPD × area; equipment $Q = (P_M/\eta_M) F_{UM} F_{LM}$
- Why sensible and latent are separated (latent is instant; sensible is delayed by thermal mass; convective vs radiative)

> 2024 Q6(b) 8 marks
>
> Distinguish between sensible and latent internal heat gains and explain why they must be treated separately in load calculations.

> 2024 Q6(d) 9 marks
>
> Describe the two methods used to estimate heat gains from lighting systems.

> 2022 Q5(b) 9 marks
>
> What is the importance of internal heat gains in building heating and cooling load calculation? What are the main components of internal heat gains?

> 2022 Q5(d) 7 marks
>
> How to calculate sensible heat gain and latent heat gain in a building?

> 2020 Q5(a) 11 marks
>
> What are the major components of internal heat gains in a building?

Full answer: ESE 4013 Study Pack, topic B5.

B7 Infiltration, ventilation and the stack effect

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 56 marks in total

STUDY THESE

- Infiltration vs ventilation (natural, mechanical); factors; leakage sites of a house
- Measurement: tracer gas decay, blower-door pressurisation; prediction: air-change (ACH) and crack method; LBL model

- Driving pressures: wind, stack (neutral pressure level), mechanical; prevention by caulking and weather-stripping

> 2024 Q7(c) 9 marks
>
> Distinguish between ventilation and infiltration. What are the typical air-leakage site of a house?

> 2022 Q6(c) 9 marks
>
> What factor influences the air infiltration through building envelope? Explain the tracer gas method to measure the infiltration rate.

> 2022 Q7(a) 6 marks
>
> What is stack effect? Explain with schematic diagram.

> 2022 Q7(c) 12 marks
>
> Distinguish between infiltration and natural ventilation. Explain air change method for calculating air infiltration rate.

> 2021 Q7(b) 8 marks
>
> How to measure the air infiltration rate in a building? Distinguish between crack method and air-change method for prediction of air infiltration.

> 2020 Q5(b) 12 marks
>
> Distinguish between infiltration and ventilation. How to prevent infiltration?

Full answer: ESE 4013 Study Pack, topic B7.

B1 Heat transfer through windows: U-factor, double pane, low-e, glazing selection

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 54 marks in total

STUDY THESE

- Window U-factor from centre-of-glass, edge and frame; single-pane resistance network (glass is only 2 %)
- Double pane: $1/U = 1/h_i + 1/h_{space} + 1/h_o$; minimise h_{space} by low-e and optimum gap
- Glazing for cold vs hot climates; colour and number of panes; low-e vs spectrally selective low-e; selection considerations

> 2024 Q5(a) 8 marks

> Explain how glazing selection affects overall occupant comfort and building performance beyond just heat gain and heat loss.

> 2024 Q5(c) 9 marks
> A homeowner in a cold northern climate complains that their living room loses heat rapidly at night through large double-pane windows. What emissivity level should be chosen for the interior-facing glass surface, and why would this be more effective than increasing the glass thickness or reducing the air gap?

> 2022 Q5(c) 10 marks
> How color and number of glazing affect the heat transfer through window? Write short note on "smart window".

> 2022 Q7(d) 4 marks
> What are the desirable characteristics of window glasses in cold location?

> 2021 Q6(a) 9 marks
> Explain the typical characteristics of window glazing in cold and hot location with schematic diagram.

> 2021 Q6(b) 9 marks
> How does color affect the heat transfer through building walls? Explain. Explain the ways to minimize the air space between the glasses of double-pane window.

> 2020 Q4(a) 5 marks
> Distinguish between 'low-e coating' and 'selective low-e'.

Full answer: ESE 4013 Study Pack, topic B1.

B12 Evaporative cooling and solar-assisted desiccant dehumidification

Must study · Very likely · 90% · Medium

Asked in 3 of 4 papers · 39 marks in total

STUDY THESE

- Direct vs indirect evaporative cooling; adiabatic cooling
- Desiccant wheel + thermal wheel + evaporative cooler; regeneration 60-90 °C by evacuated-tube collectors
- Solar-assisted desiccant with conventional AC; NZEB cooling for hot-humid climate (latent by desiccant, sensible by efficient cooling)

> 2024 Q8(c) 11 marks

> For a Net-zero Energy Building (NZEB) in a hot humid climate. Propose an appropriate cooling strategy that addresses both latent and sensible loads while minimizing energy consumption, and describe the key process steps involves.

> 2024 Q8(d) 11 marks
> Discuss about solar assisted desiccant dehumidification with air conditioning.

> 2022 Q8(b) 8 marks
> Explain the evaporative cooling system with solar-assisted desiccant dehumidification system.

> 2021 Q8(d) 9 marks
> Describe the evaporative cooling system with desiccant dehumidification.

Full answer: ESE 4013 Study Pack, topic B12.

B8 Annual energy: balance point, degree-day and bin methods

Must study · Very likely · 100% · Hard

Asked in 4 of 4 papers · 53 marks in total

STUDY THESE

- Annual consumption vs peak (design) load
- $T_{bal} = T_i - Q_{gain}/K_{tot}$; HDD, CDD; $Q_{yr} = K_{tot} \cdot DD/\eta$; T_{max} for ventilation-only cooling
- Bin method and why it beats degree-day (variable efficiency, solar, schedules); elements of annual estimate

> 2024 Q7(d) 8 marks
> Explain heating degree-day method with necessary equations.

> 2024 Q8(a) 6 marks
> Define annual energy consumption and explain why it differs from peak load calculation.

> 2022 Q7(b) 13 marks
> What is balance point temperature? Explain the degree-day method to estimate the heating and cooling load of building. Differentiate between degree-day method and bin method.

> 2021 Q7(c) 8 marks
> What are the elements should be considered while estimate the annual energy consumption of a building in design stage. Explain the Bin method.

> 2021 Q7(d) 8 marks
>
> What are meant by heating degree-day and cooling degree-day? How to determine the maximum threshold temperature to switching between air-conditioning and ventilation only?

> 2020 Q6(a) 10 marks
>
> Explain the degree-day method to calculate the annual energy consumption of building. What are its advantages over bin method?

Full answer: ESE 4013 Study Pack, topic B8.

B6 Transient heat flow: diurnal wall response, CTF, finite difference

Must study · Very likely · 100% · Hard

Asked in 4 of 4 papers · 50 marks in total

STUDY THESE

- Why loads must be transient (4 reasons); 1-D Fourier equation $\partial T/\partial t = \alpha \partial^2 T/\partial x^2$
- Diurnal temperature/heat-flow profile of a wall; heavy vs light wall (time lag, decrement)
- Explicit finite difference: interior and boundary (convection + radiation) node with Fourier and Biot numbers; CTF model

> 2024 Q7(a) 7 marks
>
> How does diurnal outdoor temperature variation affect heat transfer through different wall constructions? Compare the thermal responses of lightweight and heavyweight walls under same sol-air temperature profile.

> 2024 Q7(b) 11 marks
>
> Explain in detail how boundary nodes are treated in the finite difference method when exposed to combined convection and radiation. Derive the final rearranged equation using the biot number.

> 2022 Q2(c) 8 marks
>
> Graphically show and discuss the variation of temperature and heat flow of a wall for a typical day from mid-night to mid-night.

> 2022 Q6(a) 9 marks
>
> With the help of schematic diagram, mathematically explain the thermal transient heat transfer through wall.

> 2021 Q6(d) 8 marks

> Why building thermal load should be determined on transient condition? Explain the CTF model.

> 2020 Q5(c) 7 marks
> Why should building heat load be determined under transient condition? Explain.

Full answer: ESE 4013 Study Pack, topic B6.

B2 Solar heat gain coefficient (SHGC)

Must study · Likely · 70% · Medium

Asked in 2 of 4 papers · 20 marks in total

STUDY THESE

- SHGC = solar heat gain through window / solar radiation incident; $\tau + \rho + \alpha = 1$
- Single pane: SHGC = $\tau + \alpha U/h_o$; double pane formula
- Why "lower SHGC is always better" is wrong: climate and orientation (passive heating, daylight)

> 2024 Q5(b) 11 marks
> Define the solar heat gain coefficient (SHGc). Discuss why the statement "Lower SHGc is always preferable" is not appropriate for all climate or orientations.

> 2022 Q6(b) 9 marks
> What is meant by solar heat gain coefficient (SHGC)? How to calculate SHGC of a single pane glass window?

Full answer: ESE 4013 Study Pack, topic B2.

B4 Sol-air temperature, wall colour, and the sun-lit wall numerical

Must study · Likely · 70% · Hard

Asked in 3 of 4 papers · 48 marks in total

STUDY THESE

- Definition and derivation: $T_{sol\text{-}air} = T_{amb} + \alpha_s q_{solar}/h_o - \varepsilon\sigma(T_{amb}^4 - T_{surr}^4)/h_o$
- $Q = UA(T_{sol\text{-}air} - T_i)$; solar fraction = $(T_{sol\text{-}air} - T_{amb})/(T_{sol\text{-}air} - T_i)$
- Effect of absorptivity / colour: $\alpha_s/h_o = 0.026$ light, 0.052 dark

> 2024 Q6(a) 11 marks

What is sol-air temperature? Develop the equation of sol-air temperature considering a sun-exposed wall.

`2021` `Q5(b)` `7 marks`

What is sol-air temperature? Why sol-air temperature differs from ambient temperature?

`2021` `Q5(d)` `13 marks`

The south masonry wall of a house is made of 10-cm-thick red face brick, 10-cm-thick common brick, 19-mm-thick air space, and 13-mm-thick gypsum board, and its overall heat transfer coefficient is 1.8 $W/m^2 \cdot °C$, which include the effect of convection on both sides. The cooling system is to be sized on the basis of the heat gain at 3 PM solar time and sol-air temperature in the location at that time is 48 °C. The interior of the house is to be maintained at 22 °C and the area of the wall is 20 m^2. If the design ambient air temperature at that time is 35 °C, determine: (i) the design heat gain through the wall, (ii) the fraction of this heat gain due to solar heating.

`2020` `Q4(b)` `8 marks`

What is sol-air temperature? Explain the effect of absorptivity of the surface on sol-air temperature.

`2020` `Q4(c)` `9 marks`

How does color affect heat transfer through walls?

Full answer: ESE 4013 Study Pack, topic B4.

B13 Energy audit and building energy management economics

Good to have · Possible · 50% · Easy

Asked in 2 of 4 papers · 19 marks in total

STUDY THESE

- Components of a building energy audit (walk-through, data, analysis, measures, report)
- Economic goal: minimum life-cycle cost; payback, savings vs investment
- Energy policy, energy champion, targets, metering (good housekeeping)

`2022` `Q8(d)` `8 marks`

How to overcome the economic challenges of building by building energy management?

> 2021 Q7(a) 11 marks
>
> What are the major components of energy auditing of a building? Discuss about the economic goal of energy management in building?

Full answer: ESE 4013 Study Pack, topic B13.

B9 Below-grade heat loss: basement walls, floors, slab on grade

Good to have · Possible · 50% · Medium

Asked in 2 of 4 papers · 19 marks in total

STUDY THESE

- Ground coupling; soil surface temperature $T_{soil}(0,t) = \bar{T} + \Delta T \sin(2\pi t/8760 - \phi - \pi/2)$
- ASHRAE basement wall: $Q = U_{avg,bw} P_b h_{bw} [T_i - T_{soil}]$; radial isotherms
- Slab on grade: $Q = F_p P_{slab} (T_i - T_o)$

> 2022 Q5(a) 9 marks
>
> How to calculate heat losses from the vertical basement walls?

> 2021 Q5(a) 10 marks
>
> Explain the ASHRAE method of calculating heat losses from walls and floor of basement. Write the equation to calculate transient temperature of soil.

Full answer: ESE 4013 Study Pack, topic B9.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ACH	air changes per hour
ALT	solar altitude angle
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BEMS	building energy management system
CDD	cooling degree-days
COP	coefficient of performance

CTF	conduction transfer function
dB	decibel (sound level unit)
DF	daylight factor
EUI	energy use intensity
HDD	heating degree-days
HSA	horizontal shadow angle
HVAC	heating, ventilation and air-conditioning
IAQ	indoor air quality
IEQ	indoor environmental quality
LC	liquid crystal
LED	light-emitting diode
LEED	Leadership in Energy and Environmental Design (green building rating)
NRC	noise reduction coefficient
NZEB	net-zero energy building
PMV	predicted mean vote (comfort index)
PPD	predicted percentage dissatisfied
PV	photovoltaic (solar cells)
SC	shading coefficient
SF	solar factor
SHGC	solar heat gain coefficient
STC	sound transmission class
UHI	urban heat island
VOC	volatile organic compound
VOCs	volatile organic compounds
VSA	vertical shadow angle
VT	visible transmittance

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